

Multiple Choice (10 marks)

1. Considering 0.1 M aqueous solutions of each of the following, which solution has the lowest pH?
 - (a) Na_2CO_3
 - (b) Na_3PO_4
 - (c) Na_2S
 - (d) NaCl
2. Of the following ions, which has the smallest radius?
 - (a) K^+
 - (b) Ca^{2+}
 - (c) Sc^{3+}
 - (d) Rb^+
3. Which of the following is an n-type semiconductor?
 - (a) Silicon
 - (b) Diamond
 - (c) Silicon carbide
 - (d) Arsenic-doped silicon
4. Which of the following is always true of a spontaneous process?
 - (a) The process is exothermic.
 - (b) The process does not involve any work.
 - (c) The entropy of the system increases.
 - (d) The total entropy of the system plus surroundings increases.
5. Calculate the Q-factor for an X-band EPR cavity with a resonator bandwidth of 1.58 MHz.
 - (a) $Q = 1012$
 - (b) $Q = 2012$
 - (c) $Q = 3012$
 - (d) $Q = 6012$

True / False (10 marks)

Answer True or False

6. True or False: Each hybrid sp^3 orbital can hold a maximum of two electrons, not four.
7. True or False: A three-component mixture can only have a maximum of three phases at equilibrium.
8. True or False: SCl_2 is the least likely to act as a Lewis acid due to its lack of empty orbitals for accepting electron pairs.
9. True or False: T_2 , unlike T_1 , is sensitive to molecular motions at the Larmor frequency.
10. True or False: Fluorine (F) has a lower electron affinity than calcium (Ca) due to its high electronegativity.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the factors influencing the chemical shift in NMR spectroscopy, specifically analyzing why CH_3Cl would have a ^1H resonance approximately 1.55 kHz away from TMS in a 12.0 T magnetic field.
12. Discuss the ^{13}C NMR spectrum of 2-methylpentane, focusing on the significance of its five distinct chemical shifts and their implications for molecular structure and isomerism.

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